

Arjun Giriraj Day 16

1. What is Wireshark?
Wireshark is a tool used to monitor network traffic and inspect network packets. Its legitimate uses are network security, incidence investigation, security monitoring, protocol analysis, etc.
2. What is packet capture?
Packet capture is the process of intercepting or recording the network traffic that is travelling across network interface.
3. What is a capture interface?
The Wireshark captures packets through an active interface where traffic activity is happening.
4. Name the three main Wireshark panes.
 - Packet lists
 - Packet details
 - Packet bytes
5. What is the difference between source and destination IP?
Source IP is the address where the packet is sent from while Destination IP is the address which received the packet.
6. What does the Protocol column show?
Protocol column shows protocols like ARP, ICMP, DNS, TCP, UDP, TLS, HTTP, IPv4, IPv6.
7. What is a Wireshark display filter?
Display filter used to show the packets with specific types like protocols, dns, etc. It is shown from the already captured packets.
8. Which filter shows ICMP traffic?
Ping style traffic.
9. Which filter shows DNS traffic?
dns.
10. Which filter shows TCP traffic?
tcp.
11. What does `tcp.port == 443` display?
TCP traffic touching port 443.
12. What is DNS?
Domain name server. And in Wireshark dns refers to the packet filtering, used to show dns queries/response.
13. What is an ICMP Echo Request?

ICMP echo request is the diagnostic packet sent by the host to verify the connectivity with the destination.

14. What is an ICMP Echo Reply?

ICMP echo reply is the packet sent by destination to the host know that connection is done.

15. What is the TCP three-way handshake?

Three way handshake in the TCP is for the confirming that the connection is ready with between both systems.

16. What are SYN, SYN-ACK and ACK?

In SYN packet system asks another system for connection, SYN-ACK packet is the reply from another system for confirming that this system is ready too, ACK is the packet that system sends to confirm the connection.

17. Why is HTTPS content normally not visible as plain text?

Because HTTPS uses TLS encryption.

18. What is a PCAP/PCAPNG file?

These are the standard packets capture file format.

19. Why is Wireshark useful in incident response?

Wireshark is useful in the incident response as it allows analysts to see what happened on the wire. It reveals specific protocols, handshake patterns, and data contents.

20. Why must packet capture be performed only on authorized traffic?

Packet capture on the authorized traffic should be performed only on authorized traffic because if we perform capture on unauthorized traffic could expose sensitive data and violates strict privacy laws.

dns filter

No.	Time	Source	Destination	Protocol	Length	Info
1	6.000255	2401:4900:bfc1:eb94	2401:4900:bfc1:eb94	DNS	100	Standard query 0x2ab2 AAAA accounts.youtube.com
9	7.482916817	2401:4900:bfc1:eb94	2401:4900:bfc1:eb94	DNS	100	Standard query 0x2ab2 AAAA accounts.youtube.com
10	7.482369586	2401:4900:bfc1:eb94	2401:4900:bfc1:eb94	DNS	100	Standard query 0xc939 A accounts.youtube.com
11	7.480465464	2401:4900:bfc1:eb94	2401:4900:bfc1:eb94	DNS	95	Standard query 0x4d38 ssl.gstatic.com
12	7.610859572	2401:4900:bfc1:eb94	2401:4900:bfc1:eb94	DNS	178	Standard query response 0xc938 HTTPS accounts.youtube.com CNAME www3.l.google.com SOA ns1.google.com
13	7.61085121	2401:4900:bfc1:eb94	2401:4900:bfc1:eb94	DNS	240	Standard query response 0x2ab2 AAAA accounts.youtube.com CNAME www3.l.google.com AAAA 2404:6880:4013:807::71 AAAA 2404:6880:4013:808::71
14	7.610951279	2401:4900:bfc1:eb94	2401:4900:bfc1:eb94	DNS	224	Standard query response 0xc939 A accounts.youtube.com CNAME www3.l.google.com A 192.178.134.113 A 192.178.134.102 A 192.178.134.139
15	7.610951347	2401:4900:bfc1:eb94	2401:4900:bfc1:eb94	DNS	111	Standard query response 0x4d38 A ssl.gstatic.com A 192.178.158.94
16	7.610950950	2401:4900:bfc1:eb94	2401:4900:bfc1:eb94	DNS	97	Standard query response 0xc938 HTTPS accounts.youtube.com
17	7.615441052	2401:4900:bfc1:eb94	2401:4900:bfc1:eb94	DNS	95	Standard query 0x6312 HTTPS ssl.gstatic.com
20	7.645383834	2401:4900:bfc1:eb94	2401:4900:bfc1:eb94	DNS	147	Standard query response 0xc938 HTTPS www3.l.google.com SOA ns1.google.com
21	7.645384263	2401:4900:bfc1:eb94	2401:4900:bfc1:eb94	DNS	117	Standard query 0x6312 HTTPS ssl.gstatic.com
82	9.482373782	2401:4900:bfc1:eb94	2401:4900:bfc1:eb94	DNS	99	Standard query 0xd356 HTTPS accounts.google.com
83	9.482542623	2401:4900:bfc1:eb94	2401:4900:bfc1:eb94	DNS	99	Standard query 0x7e94 AAAA accounts.google.com
84	9.482664925	2401:4900:bfc1:eb94	2401:4900:bfc1:eb94	DNS	97	Standard query 0xc939 A accounts.google.com
87	9.508589681	2401:4900:bfc1:eb94	2401:4900:bfc1:eb94	DNS	127	Standard query response 0x7e94 AAAA accounts.google.com AAAA 2404:24080:4013:813::54
88	9.508591742	2401:4900:bfc1:eb94	2401:4900:bfc1:eb94	DNS	149	Standard query response 0xc939 A accounts.google.com
89	9.512547466	2401:4900:bfc1:eb94	2401:4900:bfc1:eb94	DNS	115	Standard query response 0x9f78 A accounts.google.com A 192.178.158.84
120	11.106324338	2401:4900:bfc1:eb94	2401:4900:bfc1:eb94	DNS	111	Standard query 0x527e HTTPS signaler-pa.clients6.google.com
121	11.065112203	2401:4900:bfc1:eb94	2401:4900:bfc1:eb94	DNS	97	Standard query 0x6956 AAAA signaler-pa.clients6.google.com
131	11.106648221	2401:4900:bfc1:eb94	2401:4900:bfc1:eb94	DNS	111	Standard query 0x4955 A signaler-pa.clients6.google.com
147	16.259992948	10.136.237.85	10.136.237.166	DNS	102	Standard query 0x527e HTTPS signaler-pa.clients6.google.com OPT
148	16.260048348	10.136.237.85	10.136.237.166	DNS	107	Standard query 0x4955 A signaler-pa.clients6.google.com
149	16.260095546	10.136.237.85	10.136.237.166	DNS	102	Standard query 0x6956 AAAA signaler-pa.clients6.google.com OPT
150	16.434480936	10.136.237.85	10.136.237.166	DNS	107	Standard query response 0x4955 A signaler-pa.clients6.google.com A 172.217.26.106
151	16.434484882	10.136.237.166	10.136.237.85	DNS	119	Standard query response 0x6956 AAAA signaler-pa.clients6.google.com AAAA 2404:6880:4092:830::208a
152	16.349815923	10.136.237.85	10.136.237.166	DNS	91	Standard query 0x504f A signaler-pa.clients6.google.com
153	16.349826952	10.136.237.85	10.136.237.166	DNS	91	Standard query 0xc657 AAAA signaler-pa.clients6.google.com
154	16.35385276	10.136.237.166	10.136.237.85	DNS	152	Standard query response 0x527e HTTPS signaler-pa.clients6.google.com SOA ns1.google.com OPT
155	16.354073713	10.136.237.85	10.136.237.166	DNS	91	Standard query 0xd088 HTTPS signaler-pa.clients6.google.com
156	16.377187208	10.136.237.166	10.136.237.85	DNS	107	Standard query response 0x4955 A signaler-pa.clients6.google.com A 172.217.26.106

[illegible]

tcp.port == 443

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	2401:4900:bfc1:eb94::	2401:4900:bfc1:eb94::	TCP	60	556038 → 443 [ACK] Seq=1 Ack=1 Win=0 Len=0 TSval=40561243 TSecr=3670951766
2	0.240576379	2401:4900:bfc1:eb94::	2401:4900:bfc1:eb94::	TCP	86	[TCP ACKed unseen segment] 443 → 56038 [ACK] Seq=1 Ack=2 Win=1030 Len=0 TSval=3670998945 TSecr=40424792
3	4.949327644	2606:4700:90c1:d8b7::	2401:4900:bfc1:eb94::	TLSv1.2	110	Application Data
4	4.949690272	2401:4900:bfc1:eb94::	2606:4700:90c1:d8b7::	TLSv1.2	114	Application Data
5	4.967680840	2606:4700:90c1:d8b7::	2401:4900:bfc1:eb94::	TCP	86	443 → 39302 [ACK] Seq=25 Ack=29 Win=18 Len=0 TSval=3843127095 TSecr=1236860459
6	7.729365950	140.82.113.26	10.136.237.85	TLSv1.2	91	Application Data
64	7.739119581	10.136.237.85	140.82.113.26	TLSv1.2	95	Application Data
73	8.620213780	140.82.113.26	10.136.237.85	TCP	66	443 → 60942 [ACK] Seq=26 Ack=30 Win=75 Len=0 TSval=1092622470 TSecr=3657357027
79	9.452818486	2606:4700:90c1:d8b7::	2401:4900:bfc1:eb94::	TLSv1.2	128	Application Data
89	9.453187505	2401:4900:bfc1:eb94::	2606:4700:90c1:d8b7::	TLSv1.2	114	Application Data
81	9.453922452	2401:4900:bfc1:eb94::	2606:4700:90c1:d8b7::	TCP	206	Application Data
85	9.492817036	2606:4700:90c1:d8b7::	2401:4900:bfc1:eb94::	TCP	86	443 → 47818 [ACK] Seq=43 Ack=29 Win=16 Len=0 TSval=1079936056 TSecr=689490277
86	9.497155364	2606:4700:90c1:d8b7::	2401:4900:bfc1:eb94::	TCP	86	443 → 47818 [ACK] Seq=43 Ack=149 Win=16 Len=0 TSval=1079936061 TSecr=689490278
122	10.886146541	2606:4700:90c1:d8b7::	2401:4900:bfc1:eb94::	TLSv1.2	110	Application Data
123	10.886517475	2401:4900:bfc1:eb94::	2606:4700:90c1:d8b7::	TLSv1.2	114	Application Data
124	11.091445196	2606:4700:90c1:d8b7::	2401:4900:bfc1:eb94::	TCP	86	443 → 51620 [ACK] Seq=25 Ack=29 Win=16 Len=0 TSval=4167105150 TSecr=3547812297
135	13.243849069	2606:4700:90c1:d8b7::	2401:4900:bfc1:eb94::	TLSv1.2	110	Application Data
136	13.244889063	2401:4900:bfc1:eb94::	2606:4700:90c1:d8b7::	TLSv1.2	114	Application Data
137	13.248542901	216.198.79.1	10.136.237.85	TLSv1.2	105	Application Data
138	13.248676696	10.136.237.85	216.198.79.1	TLSv1.2	105	Application Data
139	13.343422133	216.198.79.1	10.136.237.85	TCP	66	443 → 55524 [ACK] Seq=40 Ack=40 Win=372 Len=0 TSval=551968583 TSecr=1853093910
140	13.343422564	2606:4700:90c1:d8b7::	2401:4900:bfc1:eb94::	TCP	86	443 → 36024 [ACK] Seq=25 Ack=29 Win=18 Len=0 TSval=4244856423 TSecr=112317031
143	14.169661696	10.136.237.85	57.144.49.32	TLSv1.2	136	Application Data
144	14.366390893	57.144.49.32	10.136.237.85	TCP	66	443 → 45990 [ACK] Seq=1 Ack=71 Win=1867 Len=0 TSval=659248949 TSecr=3671075582
145	14.493437280	57.144.49.32	10.136.237.85	TLSv1.2	138	Application Data
146	14.535956543	10.136.237.85	57.144.49.32	TCP	66	45990 → 443 [ACK] Seq=71 Ack=73 Win=227 Len=0 TSval=3671075949 TSecr=659249173
196	18.427859885	2401:4900:bfc1:eb94::	2401:4900:bfc1:eb94::	TCP	86	46440 → 443 [ACK] Seq=1 Ack=1 Win=78 Len=0 TSval=1332680072 TSecr=3923484149
197	18.430000000	2401:4900:bfc1:eb94::	2401:4900:bfc1:eb94::	TCP	86	46440 → 443 [ACK] Seq=1 Ack=2 Win=1040 Len=0 TSval=1332529319 TSecr=1332580124
200	21.323780255	10.136.237.85	57.144.49.32	TLSv1.2	488	Application Data
201	21.532338942	57.144.49.32	10.136.237.85	TCP	66	443 → 45990 [ACK] Seq=73 Ack=493 Win=1866 Len=0 TSval=659250124 TSecr=3671082736
202	21.737591432	57.144.49.32	10.136.237.85	TLSv1.2	138	Application Data

Packet Details pane

```
▶ Frame 8: Packet, 100 bytes on wire (800 bits), 100 bytes captured (800 bits) on interface wlan0, id 0
▼ Ethernet II, Src: CloudNetwork_7a:da:a9 (10:6f:d9:7a:da:a9), Dst: be:b6:a4:e2:f5:64 (be:b6:a4:e2:f5:64)
  ▶ Destination: be:b6:a4:e2:f5:64 (be:b6:a4:e2:f5:64)
  ▶ Source: CloudNetwork_7a:da:a9 (10:6f:d9:7a:da:a9)
    Type: IPv6 (0x86dd)
    [Stream index: 0]
  ▶ Internet Protocol Version 6, Src: 2401:4900:bfc1:eb94:3bcc:19c5:535e:d36d, Dst: 2401:4900:bfc1:eb94::53
  ▶ User Datagram Protocol, Src Port: 52280, Dst Port: 53
  ▶ Domain Name System (query)
```

saved day16-lab.pcapng file

